Lab 4: Capacitance

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Pre-lab Questions

The capacitance of a parallel plate capacitor is given by:

$$C^* = \kappa\varepsilon_0\frac{A}{d} = \kappa C,$$

where ε_0 is the permittivity of free space, κ is the relative permittivity of the dielectric material, A is the area of each plate, and d is the distance of separation. For paper, $\kappa \approx 3.7$.

1. Calculate the capacitance for plates that are separated by $\frac{1}{4}$ inch of paper, and that have the area of an 8.5" × 11" piece of paper.

   The area of the plates is $A = (8.5\text{ in})(11\text{ in}) = (0.2159\text{ m})(0.2794\text{ m}) = 0.060\,32\,\mathrm{m}^2$.
   The separation of the plates is $d = 0.25\text{ in} = 0.00635\text{ m}$.

   The capacitance is:

   $$\begin{aligned} C^* &= \kappa\varepsilon_0\frac{A}{d} \\ &= (3.7)\left(8.85\cdot 10^{-12}\,\mathrm{F\,m^{-1}}\right)\frac{0.060\,32\,\mathrm{m}^2}{0.00635\text{ m}} \\ &\approx 3.111\cdot 10^{-10}\,\mathrm{F}. \end{aligned}$$

2. Calculate the capacitance if the area and separation are halved.

   Since $C^* \propto \frac{A}{d}$, multiplying both A and d by the same factor $\frac{1}{2}$ will multiply C^* by a factor of $\frac{1/2}{1/2} = 1$. So, the capacitance will remain the same: $C^* \approx 3.111\cdot 10^{-10}\,\mathrm{F}$.

Let there be two capacitors C_1 and C_2, each with capacitance 1 µF.

3. What is the equivalent capacitance of a series combination of C_1 and C_2?

   The potential difference across C_1 is $\Delta V_1 = q/C_1$ and the potential difference across C_2 is $\Delta V_2 = q/C_2$.
   So, the total potential difference across the series combination is:

   $$\Delta V = \Delta V_1 + \Delta V_2 = \frac{q}{C_1} + \frac{q}{C_2} = q\left(\frac{1}{C_1} + \frac{1}{C_2}\right).$$

   Assuming the equivalent capacitance of the series combination is C_{series}, then:

   $$\begin{aligned} \Delta V = \frac{\cancel{q}}{C_{\text{series}}} &= \cancel{q}\left(\frac{1}{C_1} + \frac{1}{C_2}\right) \\ \frac{1}{C_{\text{series}}} &= \frac{1}{C_1} + \frac{1}{C_2}. \end{aligned}$$

4. What is the equivalent capacitance of a parallel combination of C_1 and C_2?

   Since C_1 and C_2 are in parallel, they have the same potential difference: $\Delta V = \Delta V_1 = \Delta V_2$.
   The total charge stored in the parallel combination is:

   $$q = q_1 + q_2 = C_1\Delta V + C_2\Delta V = \left(C_1 + C_2\right)\Delta V.$$

   Assuming the equivalent capacitance of the parallel combination is C_{parallel}, then:

   $$\begin{aligned} q = C_{\text{parallel}}\Delta V &= \left(C_1 + C_2\right)\Delta V \\ C_{\text{parallel}} &= C_1 + C_2. \end{aligned}$$

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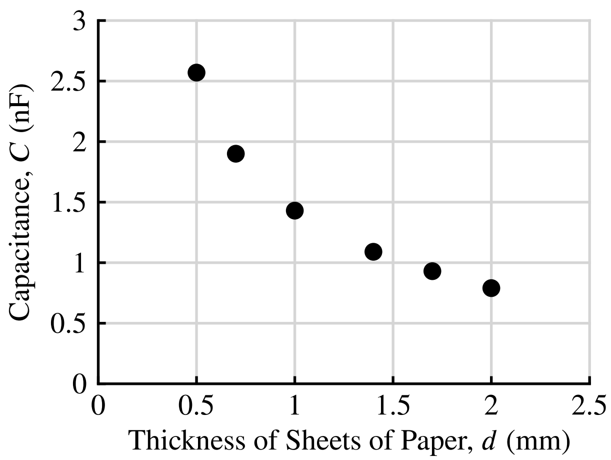
Part I: Parallel Plate with Constant Area

Measured Width	21.59 cm
Measured Length	27.94 cm
Computed Area (A)	$603.2\ \mathrm{cm}^2 = 0.06032\ \mathrm{m}^2$

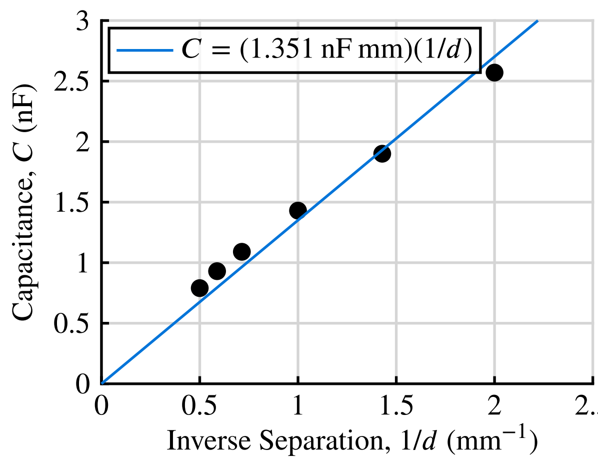
Data Collection & Analysis

Sheets of Paper	Thickness d	Measured Capacitance C
5	0.5 mm	2.57 nF
7	0.7 mm	1.90 nF
10	1.0 mm	1.43 nF
14	1.4 mm	1.09 nF
17	1.7 mm	0.93 nF
20	2.0 mm	0.79 nF

Capacitance vs. Separation of Plates



Linearized Graph



The formula for capacitance is $C = \kappa\varepsilon_0 \dfrac{A}{d} = \left(\kappa\varepsilon_0 A\right)\dfrac{1}{d}$.

The slope of the line of best fit estimates $\kappa\varepsilon_0 A$. From the C vs. $1/d$ graph:

$$\text{slope} \approx \kappa\varepsilon_0 A = 1.351\ \mathrm{nF\,mm} = 1.351 \cdot 10^{-12}\ \mathrm{F\,m}.$$

Solving for κ:

$$\kappa = \frac{\text{slope}}{\varepsilon_0 A} = \frac{1.351 \cdot 10^{-12}\ \mathrm{F\,m}}{\left(8.85 \cdot 10^{-12}\ \mathrm{F/m}\right)\left(0.06032\ \mathrm{m}^2\right)} \approx 2.530.$$

Dielectric Constant for Part I	$\kappa_1 = 2.530$

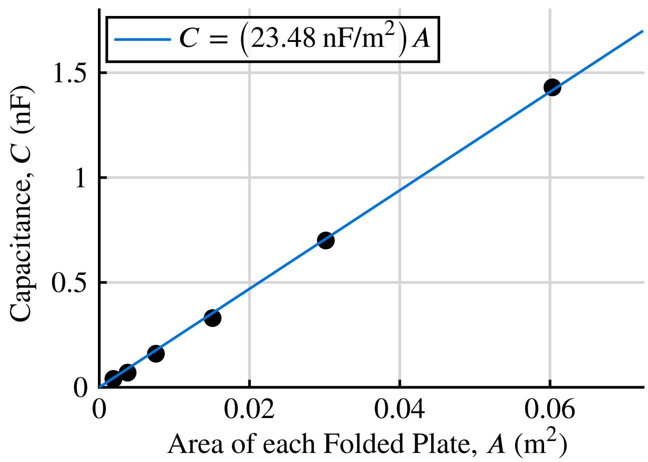
Part II: Parallel Plate with Constant Separation

The separation is fixed at 10 sheets of paper, which we measured to be $d = 1.0$ mm.

Data Collection & Analysis

Length	**Width**	**Computed Area A**	**Measured Capacitance C**
27.94 cm	21.59 cm	603.2 cm^2	1.43 nF
13.97 cm	21.59 cm	301.6 cm^2	0.70 nF
13.97 cm	10.80 cm	150.9 cm^2	0.33 nF
13.97 cm	5.400 cm	75.44 cm^2	0.16 nF
6.990 cm	5.400 cm	37.75 cm^2	0.07 nF
6.990 cm	2.700 cm	18.87 cm^2	0.04 nF

Measured Capacitance vs. Area of Folded Aluminum Sheets



Using the equation for capacitance: $C = \kappa\varepsilon_0 \dfrac{A}{d} = \dfrac{\kappa\varepsilon_0}{d} A$. The slope of the line of best fit estimates $\dfrac{\kappa\varepsilon_0}{d}$. From the C vs. A graph:

$$\text{slope} \approx 23.48\ \text{nF/m}^2 = 2.348 \cdot 10^{-8}\ \text{F/m}^2.$$

Solving for κ:

$$\kappa = \text{slope} \cdot \frac{d}{\varepsilon_0} = 2.348 \cdot 10^{-8}\ \text{F/m}^2 \cdot \frac{0.001\ \text{m}}{8.85 \cdot 10^{-12}\ \text{F/m}} \approx 2.653.$$

Percent difference:

$$\Delta\% = \frac{\left|\kappa_1 - \kappa_2\right|}{\frac{1}{2}\left(\kappa_1 + \kappa_2\right)} \cdot 100\% \approx 4.74\%.$$

Dielectric Constant for Part II	$\kappa_2 = 2.653$
Percent Difference	$\Delta\% = 4.74\%$

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Part III: Parallel Plate with Dielectric Plate

Original Area (A)	$0.06032\ \mathrm{m}^2$
Measured Thickness (d)	2.1 mm
Measured Capacitance (C)	0.80 nF

The measured capacitance with the dielectric plate ($C = 0.80\,\mathrm{nF}$) is different from Part I because the dielectric plate is made from a different material than the paper and would have a different actual dielectric constant.

Since $C = \kappa\varepsilon_0 \frac{A}{d}$, rearranging gives:

$$\kappa = \frac{Cd}{\varepsilon_0 A} = \frac{(0.80\,\mathrm{nF})(0.0021\ \mathrm{m})}{\left(8.85 \cdot 10^{-12}\ \mathrm{F/m}\right)\left(0.06032\ \mathrm{m}^2\right)} \approx 3.147.$$

The known dielectric constant for the dielectric plate is $\kappa_{\text{known}} = 3.4$. The percent error is:

$$\%\ \text{Error} = \frac{\left|\kappa - \kappa_{\text{known}}\right|}{\kappa_{\text{known}}} \cdot 100\% \approx 7.44\%.$$

Calculated Dielectric Constant	$\kappa_3 = 3.147$
Percent Error	7.44%

Part IV: Capacitors in Series and Parallel

Capacitor 1	$C_1 = 9.96\ \mu\mathrm{F}$
Capacitor 2	$C_2 = 9.79\ \mu\mathrm{F}$

Measured Capaitance, Series	$C = 4.950\ \mu\mathrm{F}$
Measured Capaitance, Parallel	$C = 19.71\ \mu\mathrm{F}$

Calculated Effective Capacitances

Series $$\frac{1}{C_{\text{series}}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{9.96\ \mu\mathrm{F}} + \frac{1}{9.79\ \mu\mathrm{F}} = \frac{1}{4.937\ \mu\mathrm{F}} \quad \Rightarrow \quad C_{\text{series}} = 4.937\ \mu\mathrm{F}.$$

Parallel $$C_{\text{parallel}} = C_1 + C_2 = 9.96\ \mu\mathrm{F} + 9.79\ \mu\mathrm{F} = 19.75\ \mu\mathrm{F}.$$

Comparison

Configuration	**Calculated**	**Measured**	**Percent Difference**
Series	4.937 μF	4.950 μF	0.260%
Parallel	19.75 μF	19.71 μF	0.203%

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Conclusion

1. What is the relationship between the capacitance and the separation of the plates?

   The capacitance is inversely proportional to the separation of the plates: $C \propto \frac{1}{d}$. As the separation increased, the capacitance decreased.

2. What is the relationship between the capacitance and the area?

   The capacitance is directly proportional to the area of each conducting aluminum sheet: $C \propto A$. As the area increased, the capacitance also increased.

3. How does the use of a dielectric between the foil sheets change the capacitance?

   The dielectric increased the capacitance by a factor κ, where $\kappa > 1$.

4. What were possible sources of errors in the experiment?

   - The aluminum foil sheets were not perfectly flat/pressed down, which may have caused uneven separation and air gaps between the foil plates and introduced variation in d. This would have also reduced the actual contact area A.
   - The two conducting aluminum sheets may not have been perfectly aligned, which would have reduced the contact area A.

5. Why does the total capacitance of two capacitors depend upon whether they are connected in series or in parallel?

   - In a parallel connection, both capacitors share the same voltage but accumulate charge independently, so the total charge stored is:

   $$q = q_1 + q_2 = \left(C_1 + C_2\right)\Delta V = C\Delta V.$$

   This gives $C = C_1 + C_2$. The effective plate area is increased (charge can accumulate on each capacitor).

   - In a series connection, the potential differences accumulate:

   $$\Delta V = \Delta V_1 + \Delta V_2 \implies \frac{q}{C} = \frac{q}{C_1} + \frac{q}{C_2}.$$

     This gives $1/C = 1/C_1 + 1/C_2$. The effective separation is increased.